



Research article

Spatial and temporal trends of surface urban heat islands in the Andean city of Cuenca, Ecuador

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ABSTRACT

This study aims to assess the long-term spatio-temporal dynamics and magnitude of the Surface Urban Heat Island (SUHI) in the intermediate, high-altitude Andean city of Cuenca, Ecuador (~2560 m a.s.l.). This research addresses a significant gap in urban climatology by analyzing the SUHI evolution over 34 years, utilizing Landsat data from 1989 to 2022. The analysis focused on the correlation between Landsat-derived Land Surface Temperature (LST) and key biophysical indices: the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Built-up Index (NDBI), which serve as proxies for characterizing surface cooling and warming mechanisms, respectively. The results confirm a strong and intensifying SUHI effect in Cuenca, characterized by a net increase of 1.28 °C in SUHI Intensity (SUHII) over the study period, rising from 4.47 °C (1989) to 5.75 °C (2022). Urban areas consistently exhibited mean temperatures 4.47–5.75 °C higher than surrounding unurbanized regions. The mechanisms driving this thermal anomaly are confirmed by strong correlations: LST exhibits a negative correlation with NDVI ($R^2 = 0.22$ to 0.59), suggesting that vegetative cover provides evaporative cooling, while the strong positive correlation with NDBI ($R^2 = 0.47$ to 0.59) indicates that the increase in impervious, thermally massive surfaces contributes significantly to SUHI intensification. These findings provide specific evidence for the magnitude and long-term trend of surface warming within a high-altitude Andean environment, underscoring the necessity of adopting climate-responsive urban planning measures centered on green infrastructure and surface material selection to mitigate the escalating SUHI effect in this context.

1. Introduction

The rapid urbanization driven by social and economic development has led to the widespread transformation of natural landscapes characterized by vegetation and permeable surfaces into urbanized and impervious surfaces. This process has garnered significant attention, especially in developing cities, where excessive urban growth has intensified environmental challenges and sharply reduced habitable spaces. According to The United Nations [1], 68% of the global population is projected to live in urban areas by 2050, posing substantial implications for local environments, urban planning, and overall quality of life. Among the most critical environmental

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consequences of urbanization is its impact on local temperature, notably through the formation of urban heat islands (UHIs). This phenomenon is defined by elevated atmospheric or surface temperatures in urban areas compared to their suburban or rural surroundings [2].

The Surface Urban Heat Island (SUHI) phenomenon is a direct consequence of the profound biophysical modifications imposed by urbanization, which fundamentally alter the local surface energy balance (SEB) [3]. The conversion of natural, vegetated, and permeable land into built environments, dominated by impervious surfaces, results in several critical changes that intensify Land Surface Temperature (LST) [4,5]. Firstly, typical urban construction materials, such as asphalt and concrete, generally exhibit a low albedo and high heat capacity [6], which maximizes the absorption and storage of solar radiation during the day, leading to delayed thermal release at night [7]. Secondly, the replacement of natural vegetation drastically curtails evapotranspiration (latent heat flux) [3,8]. This forces a higher fraction of the available net radiation to be partitioned into sensible heat flux, substantially raising the LST compared to cooler, surrounding rural areas [9]. Finally, the complex urban geometry (canyon effect) can further inhibit longwave cooling and enhance radiative absorption, increasing the thermal anomaly.

Given that the UHI phenomenon stems from fundamental modifications to the surface energy balance [10], quantifying this thermal anomaly necessitates accurate measurement [11]. Traditionally, the overall UHI effect has been evaluated using two main approaches: (i) air temperature measurements and (ii) satellite-derived LST [12,13]. While air temperature reflects atmospheric thermal comfort, measuring it is often challenging, especially in large, heterogeneous urban areas, due to the sparse spatial coverage of ground meteorological stations [8]. Consequently, research has widely shifted towards LST measurements obtained via thermal remote sensing technologies [13,14]. The thermal anomaly derived from LST is formally known as the SUHI [14]. The use of SUHI is the preferred method for analyzing the spatial patterns and intensity of urban thermal environments because satellite data provides high spatial resolution and wall-to-wall coverage [9], capturing the fine-scale thermal heterogeneity directly linked to the surface biophysical drivers [14].

Numerous studies on the SUHI effect have utilized large-scale surface temperature data from satellites such as NOAA, AVHRR, and MODIS [15]. While these datasets are valuable, their coarse spatial resolution (e.g., typically 1 km for MODIS) limits their suitability for detailed local analyses. In this context, the TM, ETM+, TIRS, and new TIRS-2 sensors aboard Landsat 5, 7, 8, and 9 provide thermal data with spatial resolutions ranging from 60 m to 120 m [16,17]. Consequently, the Landsat series has emerged as a primary data source for long-term (multi-decadal) and high-resolution SUHI analyses, offering a balance between spatial detail and temporal

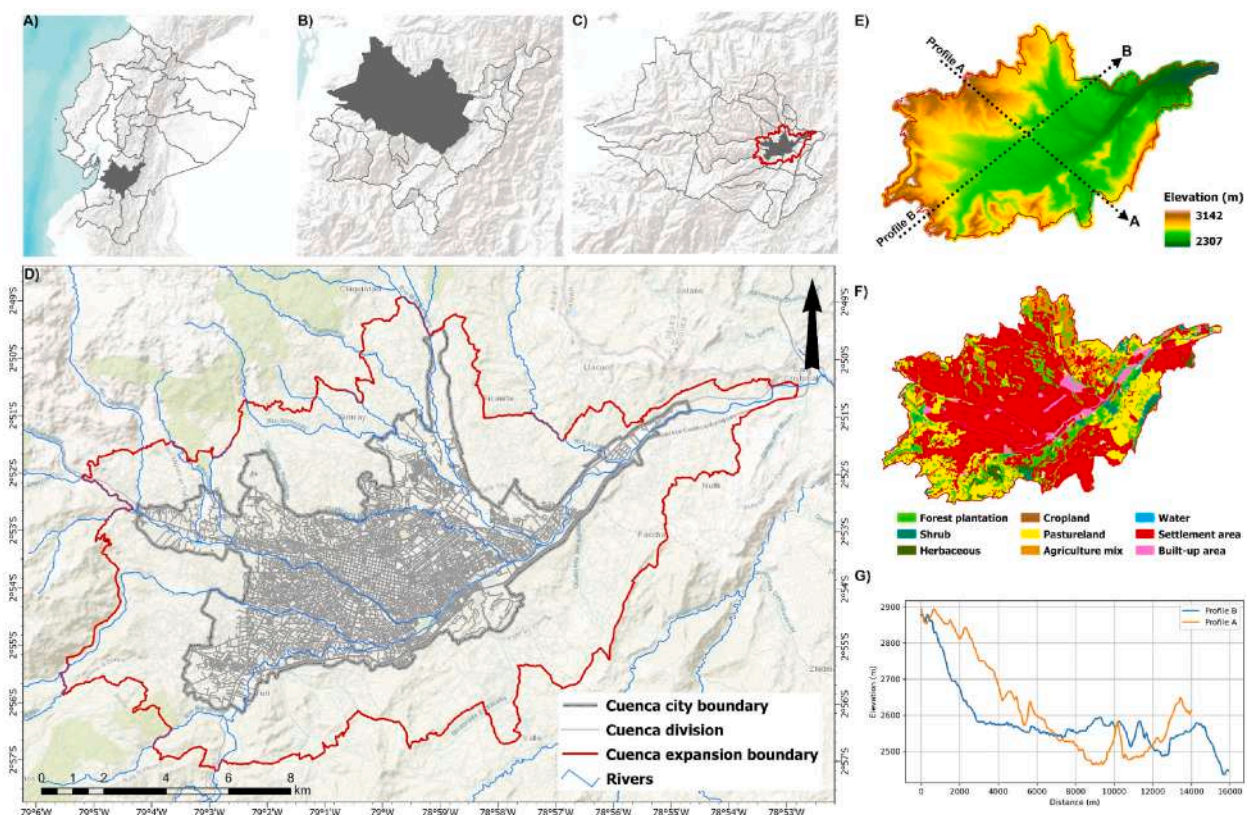


Fig. 1. The map illustrates: (A) the province of Azuay highlighted in gray, (B) the canton of Cuenca also shown in gray, (C) the city of Cuenca delineated in gray with its expansion boundary marked in red, and (D) the study area encompassing the city of Cuenca and its expansion zone. Additionally, the Digital Elevation Model (E), Land Cover (F), and Terrain Profile (G) are incorporated to provide a detailed spatial and topographic overview of the region.

coverage [17–19].

Fine-resolution Landsat data have been widely adopted for SUHI analysis globally, with key studies spanning Asia (e.g., Kuala Lumpur [20]; Beijing [18]), North America [21], Europe, and Africa [22]. In Latin America, this remote sensing approach has been applied in diverse urban areas, including Mexico City [23], Bogotá [24], and, notably, high-altitude cities in Peru [25]. Within Ecuador, satellite-based SUHI research is emerging, with studies focused on Quito [26] and Guayaquil [27]. However, existing research specific to Cuenca has primarily focused on the air temperature-based UHI using meteorological station data [28], leaving the long-term, satellite-derived surface thermal landscape unquantified.

Despite extensive global SUHI research, a critical gap persists regarding long-term spatio-temporal SUHI analysis (covering decades) in intermediate cities located within complex high-altitude Andean environments. Cities like Cuenca (~2560 m a.s.l.) present unique thermal dynamics governed by the interplay of rapid urbanization and mountainous topography. These topoclimatic conditions differ significantly from the well-documented lowland or coastal regions. Quantifying the long-term SUHI evolution under these specific altitudinal and geographical restrictions remains a fundamental gap in global urban climatology, which this study aims to address. We propose that over the 1989–2022 study period, the conversion of permeable surfaces into impervious built-up areas (evidenced by NDBI increase and NDVI decrease) has been the main spatio-temporal driver of LST rise, resulting in a quantifiable and persistent intensification of the daytime SUHI in Cuenca's Andean context. Based on this rationale, the contributions of this study are: 1) Quantification of the long-term (34-year) spatio-temporal dynamics and magnitude of the daytime SUHI in the high-altitude Andean city of Cuenca. 2) Empirical establishment of the correlation between LST, NDVI, and NDBI as proxy indicators for thermal risk assessment in this specific climate and urbanization context. 3) Provision of evidence-based parameters and thermal vulnerability maps critical for informing local sustainable urban planning and mitigation policies in high-altitude intermediate cities.

2. Materials and methods

2.1. Study area

The study area includes the urban zone of Cuenca and its expansion area, located in Cuenca Canton, Azuay Province, in southern Ecuador (Fig. 1A–D). The city of Cuenca spans 73 km², while its expansion area covers 114 km². Situated in the central-southern region of the country, the study area lies within the inter-Andean valley of the Andes Mountain range, at an altitude ranging from approximately 2560 to 3100 m above sea level (Fig. 1E–G) [28]. The predominant land cover types include forest plantations, cropland, shrub, herbaceous vegetation, pastureland, and built-up areas (Fig. 1F). As of 2022, Cuenca Canton had an estimated population of 596,101 inhabitants, with 361,524 residents (60.60%) living in the urban area [29].

Cuenca is classified as an intermediate emerging city and has experienced significant transformations over the past few decades [30]. Although located within the tropical zone, the city's high elevation results in a Cfb climate (marine west coast) according to the Köppen-Geiger classification [28]. The average annual air temperature is 16.3 °C, with a range of 10.1 °C to 21 °C [31]. The warmest months span from November to February, while the coolest months occur between June and September.

2.2. Data sources and preprocessing

Multi-decadal thermal and multispectral imagery spanning 1989–2022 were primarily sourced from the Landsat Collection 2 (C2), Level-2 (L2SP) archive via the Google Earth Engine (GEE) platform, utilizing multi-sensor data: Landsat 5 TM, 7 ETM+, 8 OLI/TIRS, and 9 OLI-2/TIRS-2. The Level-2 Science Products (L2SP) inherently include atmospheric correction for Surface Reflectance (SR) bands. LEDAPS (Landsat Ecosystem Disturbance Adaptive Processing System) was applied for Landsat 4–7 data [32], and LaSRC (Land Surface Reflectance Code) was used for Landsat 8 and 9 data [33]. The thermal infrared (TIR) bands, however, were delivered as Top-of-Atmosphere (TOA) Brightness Temperature (BT) products, which serve as the required input for the subsequent LST retrieval algorithm [20].

All reflective bands (Visible, Near-Infrared, and Short-Wave Infrared) utilized in this study possess a native spatial resolution of 30 m pixel⁻¹. While the thermal infrared (TIR) bands were originally acquired at coarser resolutions (120 m for Landsat 5 TM, 60 m for Landsat 7 ETM+, and 100 m for Landsat 8/9 TIRS), all delivered Level-2 data products were systematically resampled to a consistent 30 m pixel⁻¹ spatial resolution using the cubic convolution method by the USGS. The selected images were downloaded in GeoTIFF format and referenced to the World Geodetic System of 1984 (WGS84).

Table 1

List of datasets used for the study. The images were collected for the month of November. This month was selected as it represents a period of seasonally high temperatures and, most critically, provided the highest availability of cloud-free scenes over the study area.

Satellite	Sensor	Landsat index	Date of acquisition	Time (UTC)	Cloud cover
Landsat 5	TM	LT05_010062_19891110	1989-11-10	14:51:31	0
Landsat 7	ETM+	LE07_010062_20011103	2001-11-03	15:15:27	0
Landsat 7	ETM+	LE07_010062_20091109	2009-11-09	15:17:38	0
Landsat 8	OLI/TIRS	LC08_010062_20131128	2013-11-28	15:28:43	0
Landsat 8	OLI/TIRS	LC08_010062_20161120	2016-11-20	15:27:26	0
Landsat 9	OLI2/TIRS2	LC09_010062_20221113	2022-11-13	15:27:34	0

Stringent criteria were applied for scene selection to ensure methodological consistency for long-term analysis in this high-altitude tropical region. This included limiting acquisition to 0% cloud cover ($CLOUD_COVER = 0$), utilizing pixel-level metadata such as QA_PIXEL and QA_RADSAT for filtering saturated pixels. Furthermore, temporal standardization to the month of November was enforced across all years (1989–2022) to minimize biases originating from atmospheric interference and seasonal/phenological variability, including soil moisture and precipitation effects [19,34] inherent to the region. This temporal consistency is critical to ensure that observed long-term changes are primarily attributed to urbanization effects [18,35]. Table 1 provides detailed specifications for the satellite images utilized in this study. All datasets used are publicly available within the unified Landsat Collection 2 in the Earth Engine Data Catalog: <https://developers.google.com/earth-engine/datasets/catalog/landsat>.

The methodological decision to rely on a single, highly controlled, cloud-free Landsat image per year is fundamentally justified by the necessity of achieving robust long-term temporal consistency for a 34-year trend analysis in a cloud-prone, high-altitude Andean region [12]. The challenge of obtaining multiple high-quality images in tropical, mountainous environments due to persistent cloud cover and complex topography is well-documented in SUHI research [19,36,37]. This constrained snapshot approach, temporally standardized (e.g., to November), effectively mitigates the confounding effects of transient atmospheric conditions and substantial seasonal phenological variability inherent to the region. This ensures that the resultant long-term LST trends primarily correlate with irreversible, anthropogenic changes in biophysical surface properties driven by urban expansion [17]. Furthermore, prioritizing satellite-derived LST (SUHI) analysis is essential for capturing the fine-scale (30 m) thermal heterogeneity of the urban environment [18]. This requirement for high spatial detail is unmet by sparse meteorological station networks, which provide only limited, punctual air temperature measurements and lack the contiguous spatial coverage needed for detailed SUHI mapping [8,38].

2.3. Analytical methodology

2.3.1. Methodological framework

The methodological framework developed for the long-term spatio-temporal analysis of the SUHI phenomenon in Cuenca is structured into four sequential phases (Fig. 2): 1) Satellite Data Acquisition and Preprocessing; 2) Land Surface Variable Retrieval; 3) SUHI Quantification and Mapping; and 4) Spatio-Temporal Analysis and Validation. This structure addresses the multi-temporal and multi-spectral requirements of the study (1989–2022) and ensures the systematic assessment of LST trends and their biophysical drivers in the complex Andean environment.

2.3.2. Derivation of biophysical indices

To quantify vegetation and built-up surfaces, respectively, the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Built-up Index (NDBI) were calculated [39]. To ensure temporal consistency across the multi-sensor Landsat time series, the corresponding bands for Near-Infrared (NIR), Red (R), and Shortwave Infrared 1 (SWIR1) reflectance were used.

- Landsat 5 TM/Landsat 7 ETM+: NIR (Band 4), R (Band 3), SWIR1 (Band 5).
- Landsat 8 OLI/Landsat 9 OLI-2: NIR (Band 5), R (Band 4), SWIR1 (Band 6).

2.3.3. Land surface temperature (LST) retrieval

For this study, the Land Surface Temperature (LST) was calculated using the Planck function method proposed by Artis and

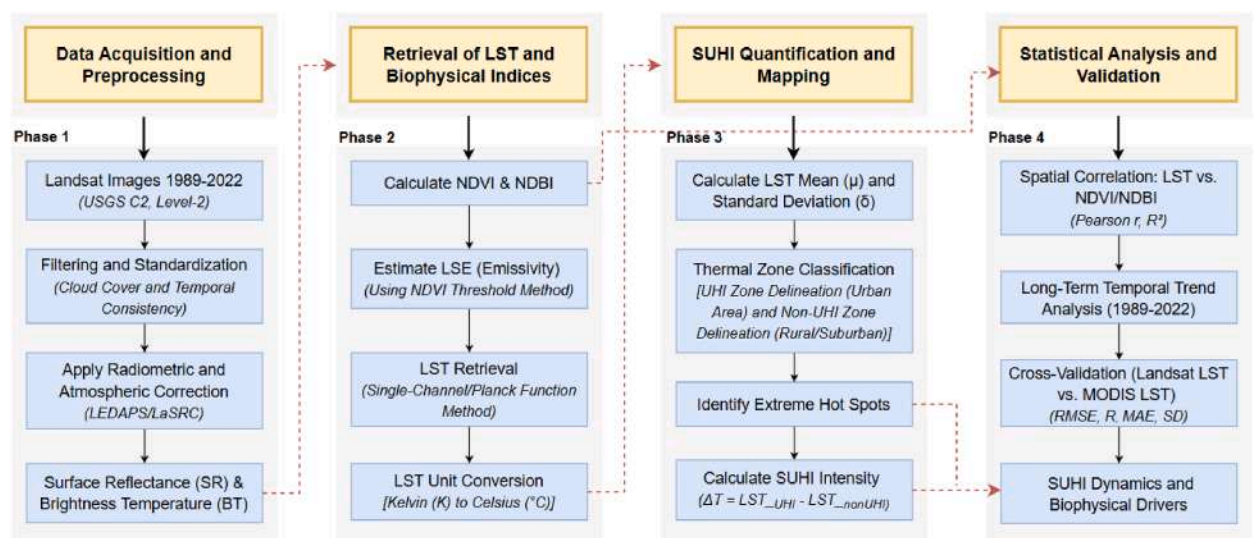


Fig. 2. Flow chart of methodology.

Carnahan [40], which has demonstrated optimal performance in various investigations [4,15]. This study utilized Landsat 8/9 Band 10 (10.6–11.19 μm) and Landsat 5/7 Band 6 (10.4–12.5 μm). The algorithm converts Brightness Temperature (T_b) to LST using the specific wavelength of emitted radiance (λ), for which values of 10.8 μm (L8/9) and 11.45 μm (L5/7) were used [15,41], and the Land Surface Emissivity (ϵ). The constant ρ was set to 1.438×10^{-2} m K [4].

Land Surface Emissivity (ϵ) was estimated using the NDVI-thresholds method [42]. This method requires calculating the Proportion of Vegetation (P_v) from NDVI. For this calculation, widely adopted standard thresholds of $NDVI_v = 0.50$ (pure vegetation) and $NDVI_s = 0.20$ (bare soil) were utilized to derive emissivity values for mixed pixels [2,42,43].

2.3.4. SUHI quantification and mapping

To identify the SUHI and non-SUHI zones in the city of Cuenca and its surrounding based on the Land Surface Temperature (LST), the distribution of LST pixels is segmented using statistical thresholds related to the mean and standard deviation of the area's LST, a method commonly utilized for characterizing thermal variations [44]. The high-temperature zones (SUHI) and the non-SUHI zones are estimated with the following criteria [44]:

$$\text{SUHI zones : } LST > \mu + 0.5 * \delta \quad (1)$$

$$\text{non - SUHI zones : } 0 < LST \leq \mu + 0.5 * \delta \quad (2)$$

Where μ = mean LST in $^{\circ}\text{C}$ and δ = standard deviation of LST. These criteria delineate areas with significantly elevated surface temperatures.

The definition of the urban-rural boundary is problematic and context-dependent in the literature [12,22]. Therefore, to quantify the Surface Urban Heat Island Intensity (SUHII), this study utilizes a statistical threshold approach (Eqs. (1) and (2)). This method was chosen because it delineates the thermal anomaly based on the LST profile itself, thus avoiding the subjectivity and errors associated with defining rigid rural boundaries [12]. This statistical approach is particularly suitable for Cuenca, as it inherently adapts to both the specific urban morphology (core and expansion zone) and the complex topoclimatic variations of the high-altitude Andean environment [28].

In this study, the SUHII has been defined as the difference between the mean LST of the classified SUHI zones and the mean LST of the non-SUHI zones [44]. This approach follows guidelines proposed by existing literature to quantify the magnitude of the thermal anomaly:

$$SUHII = \overline{LST}_{\text{SUHI zones}} - \overline{LST}_{\text{non-SUHI zones}} \quad (3)$$

Where $\overline{LST}_{\text{SUHI zones}}$ is the mean temperature of all pixels categorized in the SUHI category, and $\overline{LST}_{\text{non-SUHI zones}}$ is the mean temperature of all pixels categorized as non-SUHI.

2.3.5. Delineating the hot spots

LST maps were used to delineate the hot spots over the city of Cuenca and its expansion area in order to provide a special emphasis for continuous monitoring and territorial planning. The hot spots are very hot zones that generally occur within the SUHI and that can primarily harm vulnerable groups in the urban area of the population. The hot spots were identified with the following equation [44]:

$$LST > \mu + 2 * \delta \quad (4)$$

Where μ = mean LST in $^{\circ}\text{C}$ and δ = standard deviation of LST.

2.3.6. Validation and statistical analysis

2.3.6.1. Quantitative precision evaluation and cross-validation. Given the lack of in-situ LST measurements, a cross-validation was performed using MODIS Terra LST products (MOD11A1), following established methodologies for satellite inter-comparison. The quantitative evaluation employed statistical metrics including Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Pearson's correlation coefficient to assess the consistency between Landsat and MODIS datasets. A detailed description of the cross-validation framework is provided in [Appendix A](#).

2.3.6.2. Statistical framework for LST-NDVI and LST-NDBI correlation. The quantification of the long-term relationships between LST and its dominant biophysical drivers, the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Built-up Index (NDBI), was performed using Simple Linear Regression (SLR) models [34,35]. This approach was utilized to statistically model the universally recognized inverse thermal mechanism of vegetation ($LST \sim NDVI$) and the direct heating influence of impervious surfaces ($LST \sim NDBI$) [20,38]. The model performance and the proportion of variance explained were evaluated using the Coefficient of Determination (R^2) [17]. Concurrently, the statistical significance of the regression coefficients was determined by computing their respective p-values derived from t-tests on the slope, ensuring the statistical validity and robustness of the observed linear relationships [45].

The methodological decision to employ separate bivariate SLR models, rather than multivariate alternatives (e.g., Multiple Linear Regression – MLR, Geographically Weighted Regression – GWR, or Structural Equation Modeling – SEM), was strategically chosen to

align with the study's core objective: isolating the specific temporal forcing of NDVI and NDBI on LST. This bivariate approach was necessary to circumvent the strong, well-documented collinearity typically observed between NDVI and NDBI in urban areas, which often confounds the independent interpretation of coefficient estimates in a single global model [6,34]. Furthermore, models explicitly addressing spatial non-stationarity (e.g., GWR [46]) were not required for this analysis, as our primary objective was to quantify the aggregate long-term temporal causality and regional trend of SUHI intensification based on statistical thresholds [45].

3. Results

3.1. Spatial and temporal distribution of NDVI and NDBI

Fig. 3(A–F) shows the NDVI distribution of the study area from 1989 to 2022, and its statistical information is presented in Table 2. The highest mean NDVI of 0.47 is observed in 2013 (Fig. 3D), while the lowest of 0.24 is recorded in 2022 (Fig. 3F). There has been a steady decrease in the mean NDVI since 2013, with a sharp drop in 2022. On the other hand, the minimum NDVI fluctuated from -0.09 in 1989 to -0.56 in 2022, with the lowest value recorded in 2013. The maximum NDVI increased from 0.81 in 1989 to 0.89 in 2022, with a slight increase in the upper range over time. The low NDVI values correspond to densely urbanized areas that lack vegetation cover, particularly in the central part of Cuenca. Conversely, the expansion area to the southwest of the city exhibits high NDVI values due to its higher vegetation cover. As expected, NDVI values remain high and relatively stable in the expansion zone, indicating

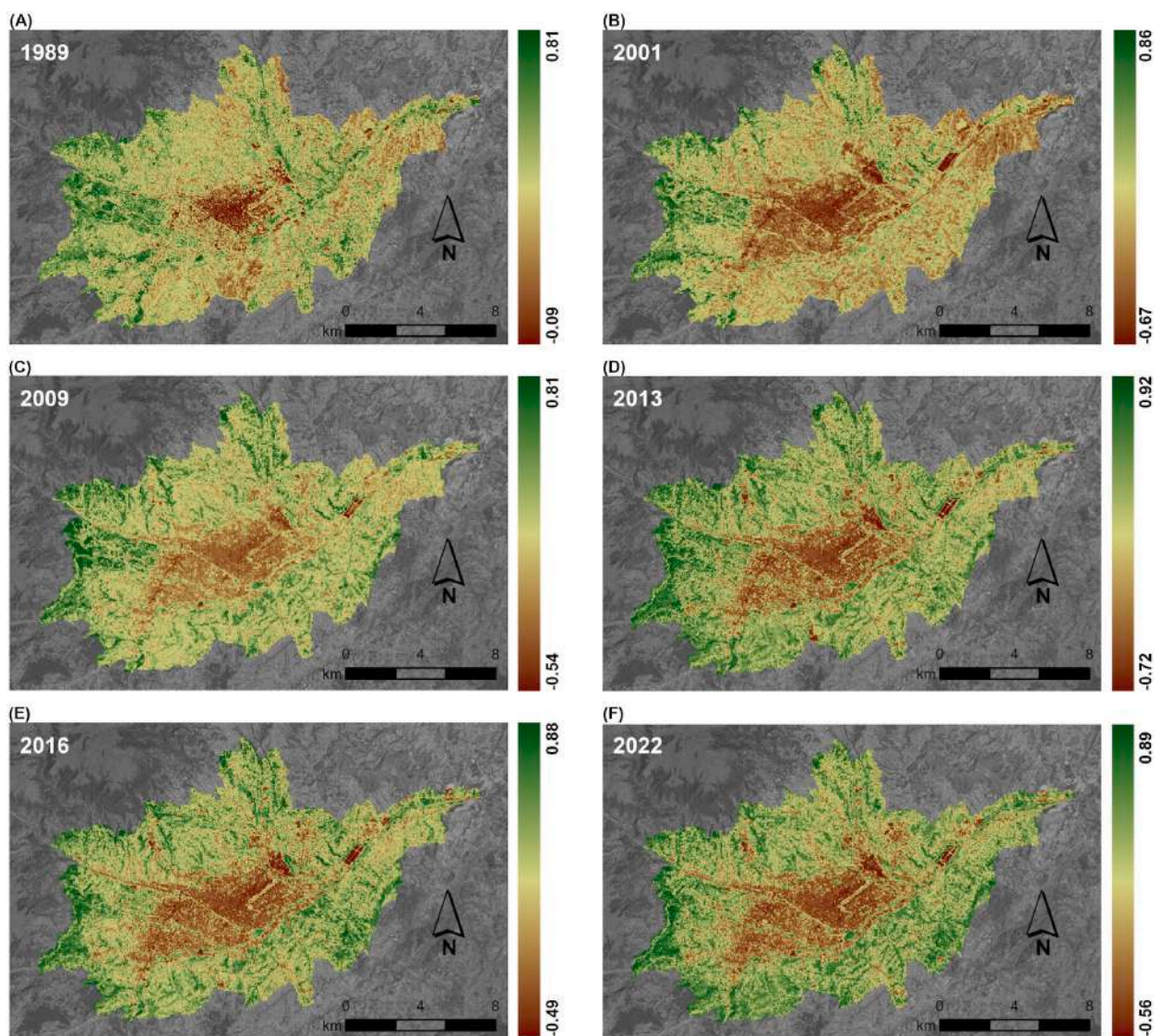


Fig. 3. Normalized Difference Vegetation Index (NDVI) map (1989–2022): A) 1989; B) 2001; C) 2009; D) 2013; E) 2016; and F) 2022.

Table 2
Descriptive statistics of LST, NDVI and NDBI for the city of Cuenca and its expansion zone (Min: Minimum; Max: Maximum; σ : Standard Deviation).

Data used	LST [°C]				NDVI				NDBI			
	Min	Max	Mean	σ	Min	Max	Mean	σ	Min	Max	Mean	σ
1989	11.09	32.49	22.90	2.90	−0.09	0.81	0.41	0.12	−0.51	0.55	−0.08	0.10
2001	10.49	44.51	32.82	3.65	−0.67	0.86	0.36	0.14	−0.87	0.72	0.01	0.10
2009	10.08	43.09	31.43	3.45	−0.54	0.81	0.34	0.13	−0.92	0.68	−0.03	0.09
2013	17.18	38.56	28.10	2.94	−0.72	0.92	0.47	0.18	−0.75	0.42	−0.09	0.12
2016	16.28	38.89	29.50	3.18	−0.49	0.88	0.43	0.17	−0.60	0.50	−0.07	0.11
2022	18.11	42.76	30.07	3.49	−0.56	0.89	0.24	0.26	−0.74	0.55	−0.05	0.10

consistent vegetation conditions in these areas over the years.

Fig. 4(A–F) depicts the distribution of NDBI from 1989 to 2022, and its statistical information is also shown in Table 2. The highest mean NDBI of 0.01 is recorded in 2001 (Fig. 4B), while the lowest mean of −0.09 is observed in 2013 (Fig. 4D). The minimum NDBI decreased from −0.51 in 1989 to −0.74 in 2022, indicating a consistent trend of decreasing NDBI values. In contrast, the maximum NDBI fluctuated between 0.42 and 0.72 across the study period, returning to approximately 0.55 in both 1989 and 2022. Generally, high NDBI values are observed in the city center, which corresponds to areas with high urbanization, while medium and lower NDBI

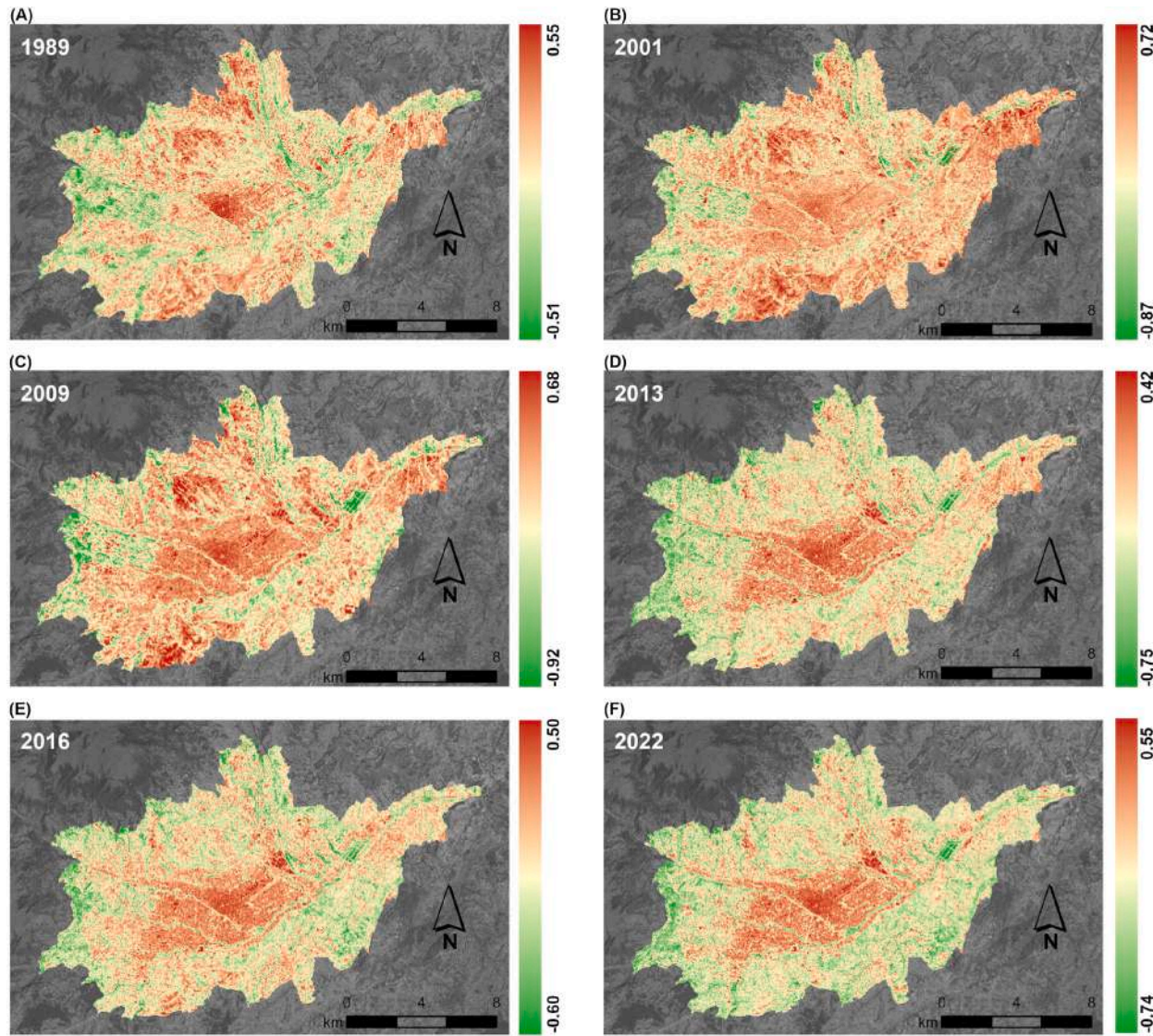


Fig. 4. Normalized Difference Built-up Index (NDBI) map (1989–2022): A) 1989; B) 2001; C) 2009; D) 2013; E) 2016; and F) 2022.

values are found on the outskirts, which typically reflect areas with lower urbanization or higher vegetation cover.

3.2. Spatial and temporal distribution of LST

Fig. 5(A–F) shows the LST of the study area from 1989 to 2022, with statistical data presented in Table 2. The highest mean LST of 32.82 °C was recorded in 2001 (Fig. 5B), while the lowest of 22.90 °C was observed in 1989 (Fig. 5A). The mean LST increased by 7.17 °C from 1989 to 2022, with the minimum temperature rising from 11.09 °C in 1989 to 18.11 °C in 2022 (Fig. 5F), and the maximum temperature increasing from 32.49 °C in 1989 to 42.76 °C in 2022. The data show fluctuating LST from 1989 to 2013, followed by a consistent increase thereafter.

The analysis of the spatial and temporal distribution of LST revealed significant variations across the city of Cuenca between 1989 and 2022. High LST values were predominantly found in urbanized areas, including built-up zones, transportation networks, and industrial regions, particularly concentrated in the city center. These areas, characterized by impervious surfaces such as asphalt and concrete, contribute to the SUHI effect, resulting in elevated temperatures compared to surrounding areas. Notably, the city's built-up zones consistently exhibited the highest temperatures throughout the study period, with maximum LST values recorded at 42.76 °C in 2022. In contrast, the presence of green spaces, including parks, gardens, rivers, and vegetated areas, has led to the creation of cooler microclimates within the urban fabric. These cooler zones were typically located on the outskirts of the city and along riverbanks. The spatial distribution of vegetation cover correlates with lower LST values, offering a natural mitigation against the SUHI effect.

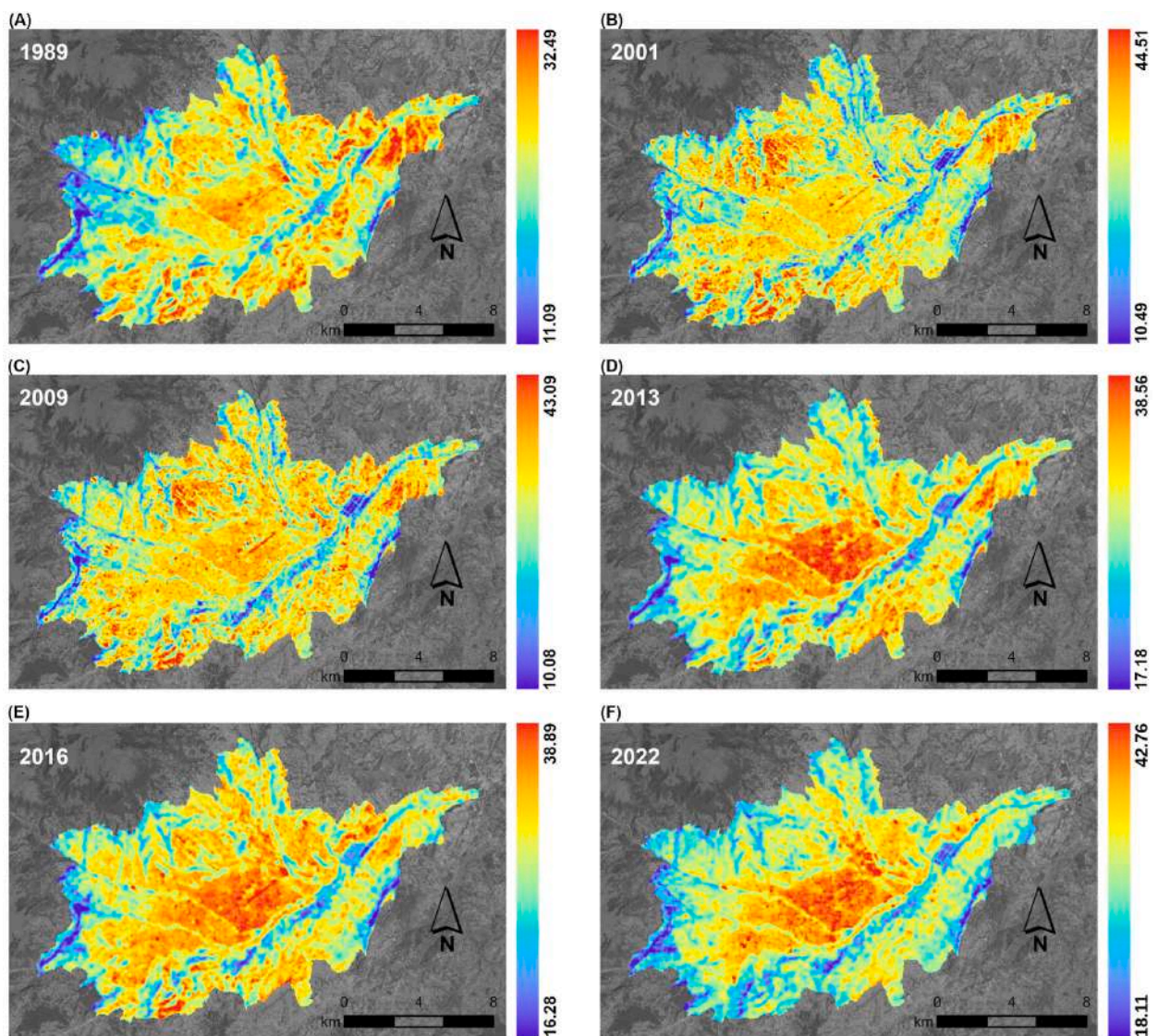


Fig. 5. Land surface temperature (LST) map (1989–2022): A) 1989; B) 2001; C) 2009; D) 2013; E) 2016; and F) 2022.

3.3. Spatial distribution of SUHI and non-SUHI

Fig. 6(A–F) illustrates the distribution of SUHI and non-SUHI zones in Cuenca for 1989 and 2022, with their statistical information summarized in Table 3. The analysis of SUHI and non-SUHI zones reveals notable thermal differences between 1989 (Figs. 6A) and 2022 (Fig. 6F). The mean LST in SUHI zones increased by 8.18 °C (from 25.83 °C to 34.01 °C), while in non-SUHI zones, it rose by 6.90 °C (from 21.36 °C to 28.26 °C). Similarly, the maximum LST in SUHI zones increased by 10.27 °C (from 32.49 °C to 42.76 °C), whereas in non-SUHI zones, it rose by 7.46 °C (from 24.35 °C to 31.81 °C).

Minimum LST values also showed a significant rise, with SUHI zones experiencing an increase of 7.46 °C (from 24.35 °C to 31.81 °C) and non-SUHI zones showing a 7.02 °C increase (from 11.09 °C to 18.11 °C). The intensity of the SUHI effect also grew, rising from 4.47 °C in 1989 to 5.75 °C in 2022, marking a net increase of 1.28 °C. These findings underscore the pronounced and consistent thermal impact of urbanization, as SUHI zones consistently exhibit higher LST values across all metrics.

3.4. Identification of hot spots

Hot spots were identified in all analyzed images and subsequently overlaid to assess their persistence and spatial extent during the period 1989–2022, as illustrated in Fig. 7. These areas are primarily located in zones characterized by intense commercial activity, high population density, minimal vegetation, dense building structures, and heavy vehicular traffic, which collectively contribute to elevated surface temperatures.

A closer examination of Fig. 7 reveals that these hot spots are predominantly found in densely urbanized areas, areas with metal roofs, and regions of bare soil. These zones lack critical elements such as water bodies and significant vegetation cover, which play a crucial role in mitigating surface temperatures through cooling mechanisms like evaporation and transpiration. These findings

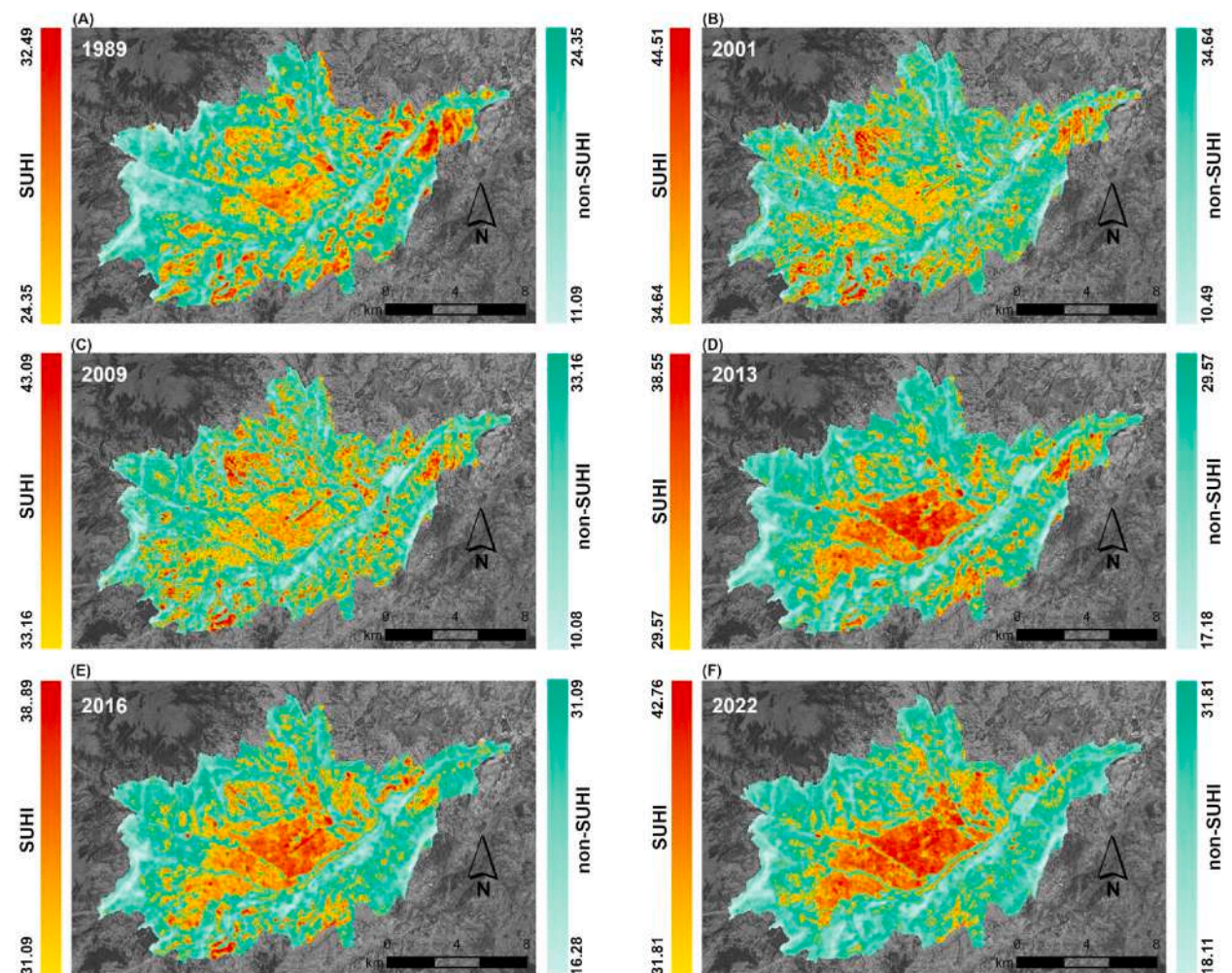


Fig. 6. Spatial extents of SUHI and non-SUHI in the city of Cuenca and its expansion area (1989–2022): A) 1989; B) 2001; C) 2009; D) 2013; E) 2016; and F) 2022.

Table 3

Variation of LST [°C] in SUHI and non-SUHI (Min: Minimum; Max: Maximum). SUHI is calculated as the difference between the mean temperature of SUHI and non-SUHI.

Data used	LST [Min]		LST [Max]		LST [Mean]		SUHI [°C]
	SUHI	Non-SUHI	SUHI	Non-SUHI	SUHI	Non-SUHI	
1989	24.35	11.09	32.49	24.35	25.83	21.36	4.47
2001	34.64	10.49	44.51	34.64	36.37	30.89	5.48
2009	33.16	10.08	43.09	33.16	34.69	29.55	5.14
2013	29.57	17.18	38.55	29.57	31.24	26.58	4.66
2016	31.09	16.28	38.89	31.09	32.67	27.80	4.87
2022	31.81	18.11	42.76	31.81	34.01	28.26	5.75

underscore the importance of green infrastructure and water features in alleviating the SUHI effect and improving urban thermal conditions.

3.5. Correlation between LST, NDVI and NDBI

Based on the regression analysis of land surface urban heat patterns between 1989 and 2022, significant correlations emerged between vegetation cover, built-up areas, and surface temperatures. The relationship between NDVI and LST exhibited consistent negative correlations ($p < 0.001$) throughout all study periods. The inverse relationship strengthened significantly over the 34-year timeframe. While R^2 values showed interannual fluctuation (with a minimum of 0.22 recorded in 2009), the overall coupling between vegetation and surface temperature increased from 0.39 (1989) to 0.59 (2022), indicating enhanced thermal mitigation (Fig. 8).

Complementarily, NDBI demonstrated strong positive correlations with LST ($p < 0.001$), with R^2 values progressing from 0.47 (1989) to 0.59 (2022). This trend suggests an intensifying relationship between built-up density and surface temperature patterns, particularly evident in recent years (Fig. 9). The parallel strengthening of both correlations underscores the dynamic interplay between urbanization, vegetation loss, and thermal conditions in the study area.

3.6. Validation of derived LST with respect to MODIS data

The overall uncertainty of the Landsat LST product was quantified using the Root Mean Square Error (RMSE), which ranged from a minimum of 3.34 °C (2022) to a maximum of 5.83 °C (2016). To provide a comprehensive assessment of error magnitude and precision, the Mean Absolute Error (MAE) and the Standard Deviation of the Differences (SD) were also computed. The MAE, representing the average magnitude of error, showed values ranging from 2.37 °C (2022) to 5.65 °C (2016). Precision, assessed directly by the SD (the scatter of the errors), varied between 1.44 °C (2016) and 2.95 °C (2022).

Furthermore, the consistency in spatial thermal patterns between the two sensors was confirmed by the Pearson's correlation coefficients (R). These demonstrated generally robust linear relationships across all dates, ranging from 0.57 (2001) to a peak of 0.82 (2016). The simultaneous reporting of RMSE, MAE, and SD establishes the reliability of the Landsat long-term record (Table A. 1). A comprehensive discussion on these validation results and their consistency with similar urban studies is provided in Appendix A.

4. Discussion

4.1. Relationship analysis of LST, NDVI and NDBI

The statistical analysis established that LST is significantly correlated with the primary biophysical indicators of urbanization: NDVI (negatively) and NDBI (positively). This dual relationship quantitatively verifies the competing thermal dynamics operating within Cuenca.

The relationship between LST and NDVI exhibited consistently negative correlations (R^2 values ranging from 0.22 to 0.59), confirming the established scientific principle that vegetative cover effectively mitigates surface temperatures [4,14,44]. This inverse correlation is rooted in biophysical cooling mechanisms, primarily evapotranspiration and shading [37,45]. Evapotranspiration facilitates the conversion of sensible heat into latent heat flux, thereby cooling the surrounding environment [19]. Furthermore, tree and canopy structures prevent direct solar radiation from reaching the surface, reducing heat absorption by impervious materials [37]. This finding, showing a moderate negative correlation strength (R^2 up to 0.59), is comparable to or stronger than values observed in other cities, such as Islamabad ($R^2 = 0.28\text{--}0.53$) [47] and Noida, India (R^2 up to 0.547) [48], validating the functional role of green infrastructure in the Andean context. This strengthening temporal trend in the LST-NDVI relationship is quantitatively substantiated by statistical shifts within the NDVI distribution (Table 2). The significant rise in the NDVI standard deviation (from 0.12 in 1989 to 0.26 in 2022) directly reflects the increasing spatial polarization and heterogeneity of the urban surface [49]. This polarization arises from the sharpening contrast between dense green spaces and expanding impervious areas [6]. Concurrently, the fluctuation toward more negative minimum NDVI values (to -0.56 in 2022) is attributed to the detection of non-vegetated extremes (e.g., shadows, water bodies), a capability amplified by the enhanced radiometric sensitivity of later Landsat OLI/TIRS instruments [50], which better captures fine-scale spectral variability across the heterogeneous urban matrix.

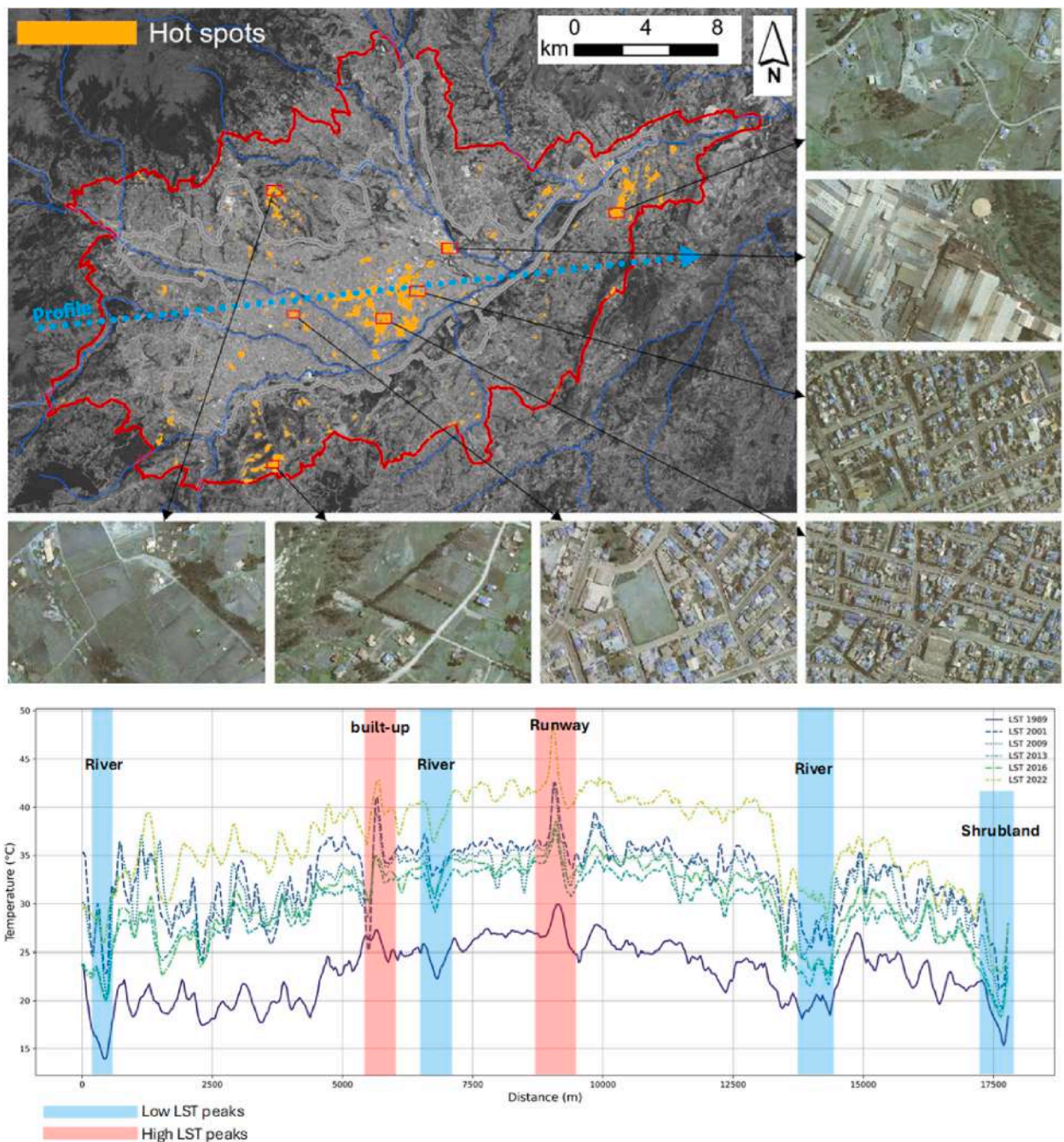


Fig. 7. Location of hot spots in the city of Cuenca and its expansion area.

Conversely, the relationship between LST and the NDBI consistently showed positive correlations (R^2 values between 0.47 and 0.59, with a strong statistical significance $p < 0.001$ throughout the period). This sustained and upward trend in the correlation coefficient confirms that the proliferation and densification of built-up areas have progressively intensified the SUHI effect over the 34-year period [14,34,44]. The fundamental reason lies in the material properties characteristic of urbanization such as asphalt, concrete, and brick, possess low surface albedo, high thermal capacity, and low moisture retention [15]. These properties promote higher sensible heat storage during the day and subsequent re-radiation, leading to elevated LST [51]. The observed correlation values (R^2 around 0.59) align strongly with similar findings in coastal Asian cities (e.g., West Bengal, $R^2 = 0.61$) [9] and notably confirm that LST is highly related to impervious surfaces [34,51]. In many urban studies, the NDBI often emerges as the dominant factor explaining LST variance due to the thermal dynamics of man-made materials [46]. The consistent and intensifying nature of the LST-NDBI relationship across the multi-decadal period provides a robust empirical foundation for the continued dominance of thermal mass effects in shaping

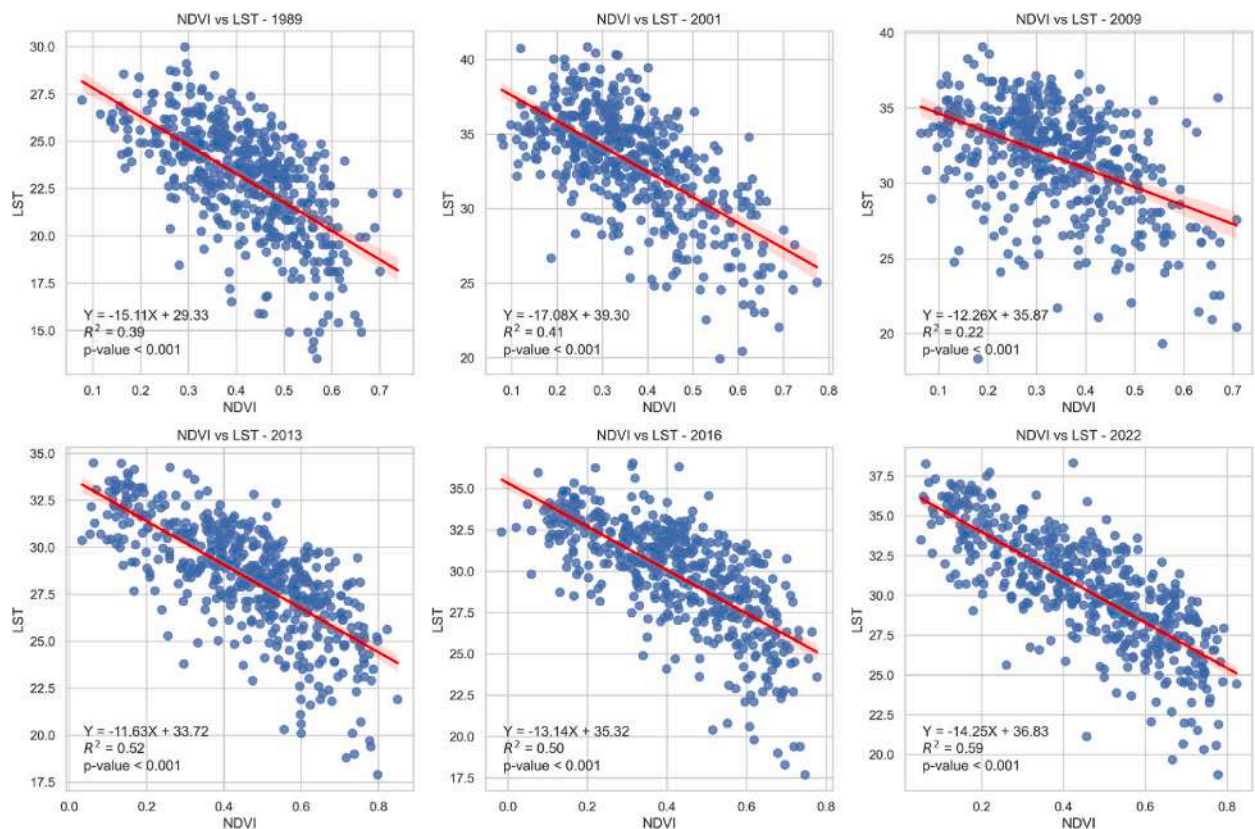


Fig. 8. Scatter plots and statistical relationship between LST and NDVI from 1989 to 2022.

Cuenca's urban thermal landscape.

The observed intensifying nature of the LST-NDVI relationship across the multi-decadal period (with R^2 values consistently rising up to 0.59) provides a robust empirical foundation for the continued dominance of thermal mass effects in shaping Cuenca's urban thermal landscape. The use of NDVI in this study inherently confirms its foundational role in biophysical modeling, particularly as the established precursor for deriving the Proportion of Vegetation (PV) [14]. The subsequent integration of PV is a crucial methodological step for accurately estimating Land Surface Emissivity (ϵ), thereby ensuring the radiometric consistency required for reliable multi-decadal LST analysis [52].

4.2. SUHI effect

The quantification of the SUHI phenomenon using satellite-derived LST data revealed a mean Surface Urban Heat Island Intensity (SUHII) in Cuenca ranging from 4.47 °C (1989) to 5.75 °C (2022), confirming the hypothesized intensification trend. This temporal increase reflects a net rise of 1.28 °C in SUHII over the 34-year study period.

These remote sensing results highlight a quantitatively higher thermal magnitude in the surface realm compared to prior localized air temperature (T_{air}) measurements in Cuenca. For instance, localized studies based on weather station data previously reported a Canopy Layer UHI (UHI-Air) value of 3.24 °C [28]. This discrepancy, where SUHI is generally more intense and spatially heterogeneous than the atmospheric UHI (T_{air}), is fundamental and stems from the underlying physical differences in measurement [44]. LST captures the radiant temperature of urban facets (e.g., roofs and roads), which are highly sensitive to the Surface Energy Balance (SEB) modifications driven by urbanization [12]. The conversion of permeable natural surfaces to impervious surfaces significantly alters the SEB [15,21]. As surfaces heat up and cool down faster than the air column, LST-derived SUHI tends to exhibit its highest magnitude during the daytime observation period [45].

The measured range of SUHII (4.47–5.75 °C) places Cuenca, an intermediate high-altitude Andean city, within the typical magnitude reported for many global urban centers. For comparative context, strong SUHII values exceeding 7 °C have been documented in tropical or rapidly developing coastal cities, such as Rio de Janeiro (7.1 °C in the last 30 years) [19] and Paço do Lumiar, Brazil [5], or higher magnitudes (up to 12 °C) observed in extreme thermal events in temperate/arid contexts [18]. Our findings, showing a strong mean SUHI effect (5.75 °C in 2022), underscore the severity of localized thermal anomalies induced by changes in surface biophysical properties [14].

Crucially, as a high-altitude Andean city (~2560 m a.s.l.), Cuenca's thermal results must be interpreted within a unique

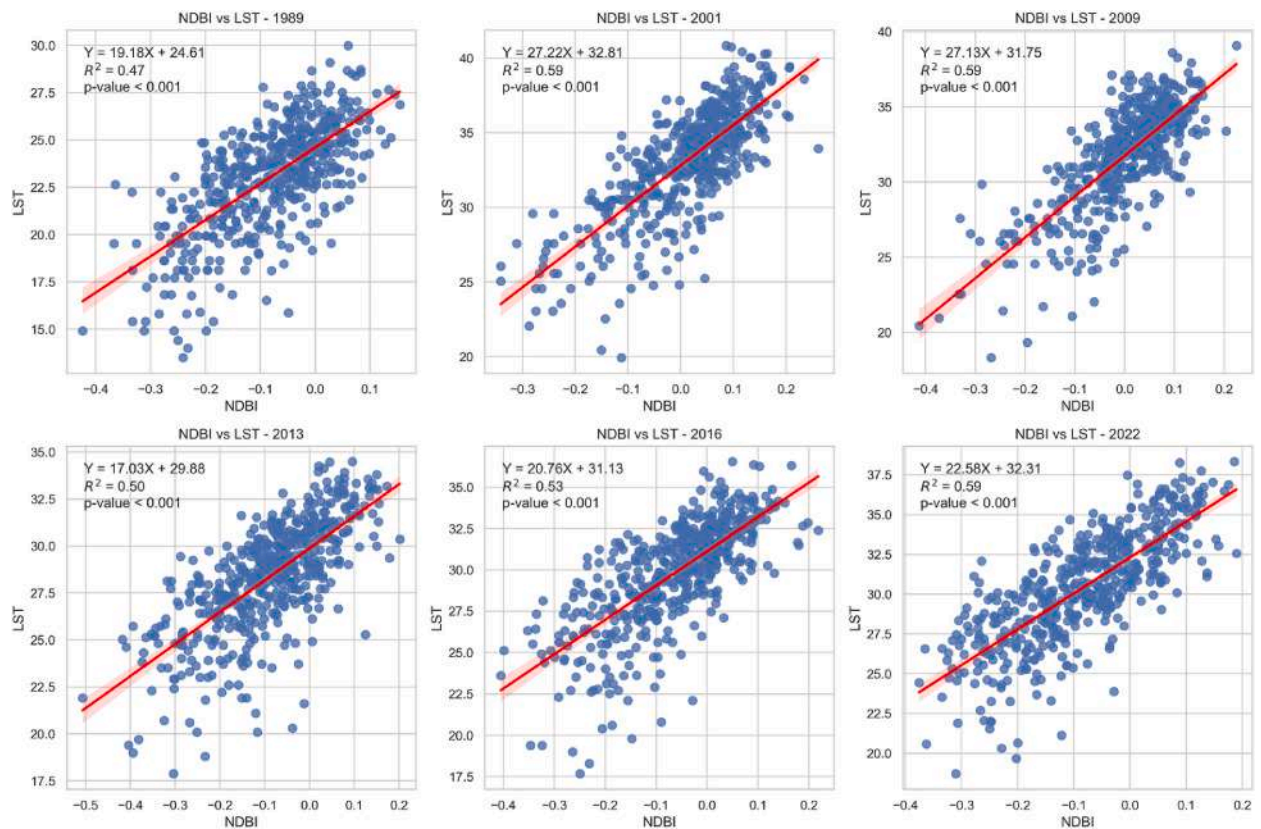


Fig. 9. Scatter plots and statistical relationship between LST and NDBI from 1989 to 2022.

topoclimatic context. The use of LST derived from remote sensing allows for continuous spatial analysis that overcomes the limitations of sparse point-source meteorological data [18]. Previous T_{air} studies in Andean contexts noted that the specific topography, where the urban area often lies at lower altitudes than surrounding rural areas, can inherently intensify the urban heat effect due to natural altitudinal thermal gradients [28]. This methodology specifically employed a statistical threshold approach to quantify SUHII (urban LST vs. non-urban LST) [44], a strategy highly suited for complex topoclimatic environments where rigid rural boundary definitions are problematic. This choice ensures that the calculated SUHII accurately represents the heating effect induced by changes in biophysical surface properties and urban expansion, thereby quantifying the long-term human modification of the thermal landscape under these specific geographical restrictions.

4.3. Limitation and future work

This long-term SUHI analysis relies on the Landsat program, the primary source for achieving a crucial balance between high spatial resolution and multi-decadal temporal coverage for thermal studies [17,18]. However, several methodological limitations must be rigorously acknowledged. The reliance on single, clear-sky images (standardized to November) introduces a significant temporal bias by failing to capture the full diurnal, seasonal, and all-weather SUHI variability [2,12,14], a common constraint in cloud-prone, high-altitude regions [3,6,37]. Furthermore, the multi-sensor time series (TM, ETM+, OLI/TIRS) introduces potential uncertainties from differences in sensor calibration and spectral response [11]. This is compounded by known artifacts, such as TIRS stray light, which can lead to absolute calibration errors and banding, particularly in Band 11 [36]. The analysis was also deliberately limited to NDVI and NDBI and did not explicitly model dynamic climatic factors (e.g., wind, humidity) [2,38] or other surface properties (e.g., soil moisture) [51], which are critical modulators of the surface energy balance [3].

Future research should address these limitations. A more comprehensive biophysical assessment is warranted, incorporating additional surface indices such as NDWI (to account for surface moisture) [35] and NDBaI (bare soil) [38], as well as SAVI (Soil Adjusted Vegetation Index), which corrects for the influence of soil brightness in areas with sparse vegetation cover [34]. To overcome temporal constraints, data fusion methodologies (e.g., combining Landsat with higher temporal resolution data like MODIS or Sentinel-3) should be employed to reconstruct diurnal and seasonal cycles [18,19,38]. This integrated dataset can then be coupled with urban microclimate models (e.g., numerical or mesoscale models) to integrate dynamic climatic variables [3] and properly incorporate the 3D thermal effects of urban morphology (such as building walls and urban canyons), which conventional satellite LST observations inherently exclude [2,12]. The community strongly encourages moving towards multi-factor driving force analysis using machine

learning methods to model the complexity of the urban thermal environment, particularly to overcome the limitations of simple linear models used here [13].

5. Conclusion

The spatio-temporal analysis spanning 34 years confirms the presence and significant intensification of the Surface Urban Heat Island (SUHI) effect in Cuenca, an intermediate high-altitude Andean city. Urban areas consistently registered LSTs approximately 5 °C higher than surrounding non-urbanized regions, reflecting a continuous shift in the local thermal balance. This trend is emphatically driven by urbanization, evidenced by the potent inverse correlation between LST and vegetation indices (NDVI) and the robust positive correlation with built-up indicators (NDBI). These correlations validate the dominant role of surface biophysical changes, specifically the expansion of impervious surfaces and the corresponding loss of green spaces, in escalating thermal risk, a pattern consistent with urbanization processes observed globally. Furthermore, the observation that hot spots are expanding into peri-urban areas underscores the pervasive nature of thermal modification beyond the traditional city core.

These findings translate directly into actionable strategies required for climate-resilient urban development. To mitigate thermal discomfort and improve environmental quality, municipal policies should prioritize comprehensive urban greening initiatives, including the expansion of urban forests and green roofs/walls in dense areas, leveraging vegetation's demonstrated cooling capacity through evapotranspiration and shading. Concurrently, resilience policies must mandate the use of cool, high-albedo materials for impervious surfaces (e.g., cool roofs and specialized pavements) in new constructions and retrofits to reduce solar absorption and sensible heat release, thereby addressing the heat retention capacity of built environments. Finally, the persistent spatial expansion of the SUHI demands that future urban planning and zoning explicitly integrate thermal vulnerability assessments (e.g., hot spot mapping), ensuring effective spatial control over LULC conversion in peri-urban and transitioning zones to promote long-term environmental sustainability and population health.

CRedit authorship contribution statement

Hermel Alvarez: Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Formal analysis. **Rasa Zalakeviciute:** Writing – review & editing, Writing – original draft, Validation. **Santiago Bonilla:** Writing – review & editing, Writing – original draft, Visualization, Supervision. **Danilo Mejía C:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Consent for publication

All authors have approved the submission.

Data availability

Data will be made available on request.

Ethical statement

Review and/or approval by an ethics committee was not needed for this study because it did not involve human participants, personal data, or animals. Informed consent was not required for this study because no human subjects were involved. This research was based exclusively on publicly available satellite remote sensing data (Landsat 5/7/8/9 and MODIS imagery).

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Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.heliyon.2026.e45397>.

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